

Thanh Nguyen Canh

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RESEARCH INTERESTS

Simultaneous Localization and Mapping (SLAM), Semantic SLAM, Active SLAM, Lifelong SLAM, Probabilistic Learning, Continual Learning, Robot Perception, Environment Representation, Motion Planning, Humanoid Robotics, Human-Robot Interaction, Multisensor Fusion and Control, UAVs.

EDUCATION

Japan Advanced Institute of Science and Technology <i>Ph. D. in Information Science, Ishikawa, Japan</i>	September 2027 - expected <i>Current GPA: 3.00/3.0</i>
Japan Advanced Institute of Science and Technology <i>M.S. in Information Science, Ishikawa, Japan</i>	September 2024 <i>GPA: 2.98/3.0, Thesis Score: 100/100</i>
VNU-University of Engineering and Technology <i>B.S.Eng in Robotics Engineering, Hanoi, Vietnam</i>	September 2022 <i>GPA: 3.77/4.0 (Top 1%), Thesis Score: 9.7/10</i>

EXPERIENCE

Visiting Researcher <i>University of Genoa, Italy</i> Advisor: Prof. Antonio Sgorbissa , RICE Lab , DIBRIS Working on the SOLARIS, HORIZON-CL5-2023-D3-02-13 project, creating drone teams for optimal monitoring of the panels in order to detect defects and faults.	2026
Teaching Assistant <i>School of Information Science, Japan Advanced Institute of Science and Technology</i> Working on slides, lectures, tutorial preparation	2024 – present
Research Assistant <i>School of Information Science, Japan Advanced Institute of Science and Technology</i> Advisor: Prof. Nak Young Chong , School of Information Science Working on Active semantic simultaneous localization and mapping for UAVs (ROS, C++, Python, PyTorch, UAVs)	2023 – present
Lecturer <i>VNU-University of Engineering and Technology</i> Working on lectures, research	2022 – Present
Research Assistant <i>Department of Robotics Engineering, VNU-University of Engineering and Technology</i> Advisor: Prof. Xiem HoangVan , Department of Robotics Engineering Working on simultaneous localization and mapping, obstacle avoidance, navigation, calibration, etc. for robotics (ROS, C++, Python, PyTorch, TensorFlow, mobile robot, dual-arm)	2019 – 2023

PROJECTS

Delibot - Humanoid Robotics <i>C++/Python, ROS, MoveIt/Gazebo, Dual-arm service robot, ground robot</i> - Optimal design and fabrication of frame structure for dual-arm service robot. <i>Papers: [J3], [C12], [P9]</i> 3D localization using 2D estimates for Robot Vision system <i>Papers: [J1], [C13]</i> - Obstacle avoidance using multi-sensor fusion. <i>Papers: [C8], [C11]</i>	2020 – 2024
Semantic Active SLAM <i>Python/C++, ROS, UAVs</i> - Real-time Semantic-Aware Simultaneous Localization and Mapping system for Unmanned Aerial Vehicles. <i>Papers: [C3], [C4], [C10], [C11], [J5]</i> - Improve localization quality based on data association method. <i>Papers: [P5, P3]</i> - Robust Visual SLAM in Dynamic Environment. <i>Paper: [J4], [C15], [C10]</i> - Autonomous Exploration and Active SLAM. <i>Paper: [P1]</i> - Gaussian Splatting for SLAM. <i>Paper: [P7]</i>	2022 – Present

- Multi-Robot Collaboration and Humanoid Perception based on Deep Learning. *Paper: [P2], [C12]*
- Humanoid Robot Interaction based on Vision Language Model/Large Language Model. *Papers: [P3], [P4], [C9]*
- Improve navigation performance in crowded environments and Semantic-aware path planning. *Papers: [C8], [C7]*

PUBLICATIONS

Preprints

- [P19] K. H. Nguyen, **T. N. Canh**, X. HoangVan, “*Uncertainty-Guided 2:4 Dynamic Sparse Evidential Learning for Rare-Event Screening*”, Association for the Advancement of Artificial Intelligence (AAAI), 2027 [website][code]
- [P18] K. H. Nguyen, **T. N. Canh**, X. HoangVan, “*Multi-head Evidential Deep Learning with Adaptive Prior Weight Fusion*”, Association for the Advancement of Artificial Intelligence (AAAI), 2027 [website][code]
- [P17] T.T. Tran, **T. N. Canh**, X. HoangVan, N. Y. Chong, “*ROCC: Reliability-Gated Obstruction Reasoning with Certified DAG for Clutter Grasping*”, IEEE International Conference on Robotics and Automation (ICRA), 2027 [website][code]
- [P16] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*From Semantic to Affordances: Hierarchical Evidential Semantic-Functional Mapping*”, IEEE Transaction on Robotics and Automation Letters (TR-O), 2026 [website][code]
- [P15] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*Compliance-Aware Reinforcement Learning for Active Semantic SLAM with a Soft Aerial Robot*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P14] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*Contact-Aware Active SLAM with a Soft UAV: Exploiting Compliance for Frontier Goal Selection*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P13] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*Uncertainty-Aware Open-Vocabulary Semantic Gaussian Splatting for Robust Mapping in Dynamic Scenes*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P12] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*HESFM-OV: Hierarchical Evidential Semantic-Functional Mapping with Open Vocabulary for Indoor UAV Navigation*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P11] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*SeLiSLAM: Semantic Lifelong Simultaneous Localization and Mapping for UAVs via Continual Learning, Open-Vocabulary Ephemerality, and Object Re-Identification*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P10] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*Open-World Semantic SLAM: A Survey on Representations, Uncertainty, and Lifelong Adaptation*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P9] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]
- [P8] H. Zhang, **T. N. Canh**, C. Li, Z. Gao, X. Jiang, N. Y. Chong, “*Con-DSO: Learning Short-Horizon Consistency Priors for RGB-D Direct Sparse Odometry*”, IEEE Transactions on Multimedia, 2026 [website][code]
- [P7] D. M. Do, V. B. Nguyen, **T. N. Canh**, M. T. Thai, X. HoangVan, “*Monocular vSLAM for Colonoscopy: Recent Advances, Benchmarking, and Future Perspectives*”, Robotics and Automation System, 2026 [website][code]
- [P6] D. M. Do, **T. N. Canh**, N. M. Do, M. T. Thai, X. HoangVan, “*Path-Aware Frontier Selection for UAV-Based Autonomous 3D Exploration*”, International Conference on Control, Automation, Robotics and Vision (ICARCV 2026) [website][code]
- [P5] M. L. Pham, D. T. Duong, X. HoangVan, **T. N. Canh**, “*Efficient Continuous Semantic Mapping based on Spatio-Temporal Awareness*”, International Conference on Control, Automation, Robotics and Vision (ICARCV 2026) [website][code]
- [P4] D.T. Nguyen, N. M. Do, X. HoangVan, **T. N. Canh**, “*Semantic-Aware Autonomous Exploration for UAVs in Unknown Indoor Environments*”, International Conference on Control, Automation, Robotics and Vision (ICARCV 2026) [website][code]
- [P3] **T. N. Canh***, P. N. Xuan*, H. Nguyen, N. Y. Chong, X. HoangVan, “*A Survey on Collaborative SLAM with 3D Gaussian Splatting*”, (Autonomous Robots 2026) (under review) [website][code]
- [P2] **T. N. Canh**, T. T. Tran, X. HoangVan, N. Y. Chong, “*Human-to-Robot Interaction: Learning from Video Demonstration for Robot Imitation*”, Autonomous Robots, 2026 (Submitted) [website][code]
- [P1] **T. N. Canh**, H. Zhang, X. HoangVan, N. Y. Chong, “*HELIX: Hierarchical Evidential Loop-closure and Information eXploration for Active Semantic SLAM in GPS-Denied Environments*”, IEEE Robotics and Automation Letters (RA-L), 2026 [website][code]

Journals

- [J8] **T. N. Canh**, N. Y. Chong, “*Bayesian Probabilistic Data Association via Gaussian Mixture Models for Semantic SLAM*”, IEEE Robotics and Automation Letters (RA-L), 2026 [paper] [website] [code]
- [J7] X. HoangVan, L. H. Luong, **T. N. Canh**, “*E2DSR: Edge-Enhanced Representation for Deep Super-Resolution in Machine Vision Applications*”, (REV-JEC 2025) [paper] [website] [code]
- [J6] **T. N. Canh**, A. P. Tuan X. HoangVan, “*Design of Deep Reinforcement Learning Approach for Traffic Signal Control at Three-way Crossroads*”, Public Transport, 2024 (accepted) [paper] [website] [code]
- [J5] **T. N. Canh**, X. HoangVan, N. Y. Chong, “*Semantic Visual Simultaneous Localization and Mapping: A Survey on State of the Art, Challenges and Future Directions*”, Robotics and Automation System, 2026 [paper] [website][code]
- [J4] H. Zhang, C. Li, **T. N. Canh**, N. Y. Chong, “*SR-SLAM: Scene Reliability-Based RGB-D SLAM in Diverse Environments*”, Robotics and Automation System(RAS), 2025 [paper] [website][code]
- [J3] **T. N. Canh**, S. T. Duc, H. N.The, T. H. Dao, X. HoangVan, “*Optimal Design and Fabrication of Frame Structure for Dual-Arm Service Robots: An Effective Approach for Human-Robot Interaction*”, Engineering Science and Technology, an International Journal (JESTECH), 2024 [paper] [paper] [code]
- [J2] **T. N. Canh**, D. M. Bui, X. HoangVan, “*ESRPCB: an Edge guided Super - Resolution model and Ensemble learning for tiny Printed Circuit Board Defect detection*”, Engineering Applications of Artificial Intelligence, 2025 [paper] [website] [code]
- [J1] **T. N. Canh**, D. T. Ngoc, X. HoangVan, “*M-Calib: A Monocular 3D Object Localization using 2D Estimates for Industrial Robot Vision System*”, Journal of Automation, Mobile Robotics and Intelligent Systems, 2025 [website][code]

Conferences

- [C17] **T. N. Canh**, T.T. Tran, H. Zhang, Z. Gao, X. HoangVan, N. Y. Chong “*Decoupled Object-Centric Video Understanding for Generating Robotic Manipulation Commands*”, (ICCAS 2026) [website][code]
- [C17] **T. N. Canh**, T.T. Viet, G. H. Uong, P. V. Dinh, T. V. T. Nguyen, X. HoangVan, N. Y. Chong “*OWaveSync: Constrained Wavefront Optimization for Synchronized Co-Speech Gestures in Humanoid Robots*”, (ICCAS 2026) [website][code]
- [C17] **T. N. Canh**, T.T. Viet, P. V. Dinh, X. HoangVan, N. Y. Chong “*OSDAG: Online Scheduling for Efficient Multi-Robot Collaboration*”, (ICCAS 2026) [website][code]
- [C16] T.T. Tran, **T. N. Canh**, N. Y. Chong, X. HoangVan “*Hybrid TD3: Overestimation Bias Analysis and Stable Policy Optimization for Hybrid Action Space*”, (CASE 2026) [website][code]
- [C15] H. Zhang, **T. N. Canh**, C. Li, R. Yang, Y. Ji, N. Y. Chong, “*IL-SLAM: Intelligent Line-assisted SLAM Based on Feature Awareness for Dynamic Environments*”, (IRC 2025) [website][code]
- [C14] T. T. Viet, P. D. Van, H. U. Gia, S. T. Duc, **T. N. Canh**, X. HoangVan, “*Development of a Humanoid Robot Prototype and Gesture and Voice-Based Interaction Approach*”, International Conference on Advances in Information and Communication Technology (RIVF 2025) [website][code]
- [C13] **T. N. Canh**, Q. M. Trinh, T-V Dang, P. X. Tan, X. HoangVan, “*Refined 3D Object Localization with Monocular Camera using Depth Estimation and Geometric Refinement*”, International Conference on Interactive Collaborative Robotics (ICR 2025) [website][code]
- [C12] **T. N. Canh**, T. T. Viet, S. T. Duc, H. N. The, T. H. Dao, V-H. Nguyen, X. HoangVan, “*Efficient Human-Robot Interaction via Deep Perception and flexible Motion Planning*”, International Conference on Interactive Collaborative Robotics (ICR 2025) [website][code]
- [C11] **T. N. Canh**, B. N. Quoc, H. Zhang, B. R. Veeraiah, X. HoangVan, N. Y. Chong, “*IRAF-SLAM: An Illumination-Robust and Adaptive Feature-Culling Front-End for Visual SLAM in Challenging Environments*”, European Conference on Mobile Robots, 2025 [website] (Accepted) [code]
- [C10] H. Zhang, **T. N. Canh**, C. Li, N. Y. Chong, “*Adaptive Prior Scene-Object SLAM for Dynamic Environments*”, The 2025 IEEE International Conference on Real-time Computing and Robotics (RCAR), 2025 (Accepted) [website][code]
- [C9] **T. N. Canh**, K. HoangVan, X. HoangVan, “*Context-aware LLM-based Human-Robot Interaction*”, International Conference on Intelligent Systems & Networks (ICISN), 2025 (Accepted) [website][code]
- [C8] **T. N. Canh**, H.-H. Ngo, X. HoangVan, N. Y. Chong, “*Toward Integrating Semantic-aware Path Planning and Reliable Localization for UAV Operations*”, The 24th International Conference on Control, Automation and Systems (ICCAS), 2024 [website] [code]
- [C7] **T. N. Canh**, X. HoangVan, N. Y. Chong, “*Enhancing Social Robot Navigation with Integrated Motion Prediction and Trajectory Planning in Dynamic Human Environments*”, The 24th International Conference on Control, Automation and Systems (ICCAS), 2024 [website][code]

- [C6] **T. N. Canh**, M. DoNgoc, T. N. Quang and H. B. Thanh, X. HoangVan, “*Underwater Image Enhancement for Depth Estimation via Various Image Processing Techniques*”, 2024 International Conference on System Science and Engineering (ICSSE), 2024 [[website](#)][[code](#)]
- [C5] M. D. Duc, **T. N. Canh**, M. DoNgoc, X. HoangVan, “*Fusion LiDAR-Inertial-Encoder data for High-Accuracy SLAM*”, 2024 International Conference on Mechatronic, Automobile, and Environment Engineering (ICMAEE), 2024 [[website](#)][[code](#)]
- [C4] **T. N. Canh**, V. Nguyen, X. HoangVan, A. Elibol, N. Y. Chong, “*S3M: Semantic Segmentation Sparse Mapping for UAVs with RGB-D Camera*”, IEEE/SICE International Symposium on System Integration (SII), 2024 [[website](#)][[code](#)]
- [C3] **T. N. Canh**, A. Elibol, N. Y. Chong, X. HoangVan, “*Object-Oriented Semantic Mapping for Reliable UAVs Navigation*”, IEEE International Conference on Control, Automation and Information Sciences (ICCAIS), 2023 [[website](#)][[code](#)]
- [C2] **T. N. Canh**, X. HoangVan, “*Machine Learning-Based Malicious Vehicle Detection for Security Threats and Attacks in Vehicle Ad-hoc Network (VANET) Communications*”, IEEE International Conference on Research, Innovation and Vision for the Future, 2023 [[website](#)][[code](#)]
- [C1] **T. N. Canh**, T. S. Nguyen, C. H. Quach, X. HoangVan, M. D. Phung, “*Multisensor Data Fusion for Reliable Obstacle Avoidance*”, IEEE International Conference on Control, Automation and Information Sciences (ICCAIS), 2022 [[website](#)][[code](#)]

TECHNICAL SKILLS

Programming: C++/Python/Matlab, HTML/CSS, L^AT_EX
Operating Systems: Linux, ROS/ROS2
Libraries and Toolbox: Pytorch, Pybullet, Gazebo/Habitat/iGibson/Unity/Rviz
Robot Platform: Dual-arm Humanoid services robot, Turtlebot3, PX4 Quadrotors

HONORS AND AWARDS

JSPS Scholarship , Japan Society for the Promotion of Science	2027-2029
JST SPRING Scholarship , Japan Science and Technology	2025-2027
JAIST Doctoral Research Fellowship (DRF) , Special Type	2024 – 2025
KDDI Scholarship , KDDI Foundation	2024-2025
Monbukagakusho Honors Scholarship	2023 – 2024
RIVF Best paper award , IEEE International Conference on Research, Innovation and Vision for the Future	2023
VinIF Scholarship for Master Programmer , Vingroup Innovation Foundation	2022-2024
Instruction students to win Second place, Students Research Competition , VNU-UET	2023
Best Student Thesis Award , VNU-University of Engineering and Technology	2022
Vallet Scholarship , Vallet Foundation	2021-2022
REV-ECIT Best paper award , Radio and Electronics Association of Vietnam	2022
First place, Students Research Competition , VNU-University of Engineering and Technology	2021

TEACHING EXPERIENCE

Teaching Assistant, System Optimization, Japan Advanced Institute of Engineering and Technology
Teaching Assistant, Discrete Signal Processing, Japan Advanced Institute of Engineering and Technology
Teaching Assistant, Fundamentals of Programming, Japan Advanced Institute of Engineering and Technology
Teaching Assistant, Robotics, Japan Advanced Institute of Engineering and Technology
Teaching Assistant, Discrete Signal Processing, Japan Advanced Institute of Engineering and Technology
Teaching Assistant, Fundamentals of Programming, Japan Advanced Institute of Engineering and Technology
Lecturer, Programming robot with ROS, VNU-University of Engineering and Technology
Lecturer, Introduction to Human Machine Interface, VNU-University of Engineering and Technology
Lecturer, Robotic Control, VNU-University of Engineering and Technology
Lecturer, Mechanical Drawing, VNU-University of Engineering and Technology
Lecturer, Electronics Engineering Practice, VNU-University of Engineering and Technology

LEADERSHIP AND MENTORING

Master students:

- | | |
|-----------------------------|--------------|
| • Pham Thi My Le (VNU-UET) | 2026-present |
| • Quoc Bao Nguyen (VNU-UET) | 2025-present |
| • Manh Do Duc (VNU-UET) | 2025-present |
| • Hoang Ngo Huy (HUST) | 2025-2026 |

Undergraduate students:

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| • Nguyen Duc Khanh Huyen (VNU-UET) | 2024-present |
| • Duong Vu Viet Thang (VNU-UET) | 2024-present |
| • Dinh Mang Quan (VNU-UET) | 2024-present |
| • Nguyen Van Bang (VNU-UET) | 2024-present |
| • Tuan Thanh Tran (VNU-UET) | 2024-present |
| • Dang Minh Bui(VNU-UET) | 2024-2025 |
| • Manh Do Duc (VNU-UET) | 2022-2024 |
| • Hoang Ngo Huy (VNU-UET) | 2022-2024 |
| • Quan Nguyen (VNU-UET) | 2022-2024 |
| • Kien Hoang (VNU-UET) | 2022-2024 |

PROFESSIONAL MEMBERSHIP & ACTIVITIES

Student Representative , IEEE Robotics and Automation Society	2026-2027
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AWIST 2025 Committee , The 13 th ASEAN Workshop on Information Science and Technology	2025
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Reviewer:

- **IEEE**: IEEE Robotics and Automation Letters, IEEE Transactions on Robotics, IEEE Transactions on Automation Science and Engineering.
- **Elsevier**: Knowledge-Based Systems, Engineering Applications of Artificial Intelligence, ISA transactions, Expert systems with applications, Engineering Science and Technology, an International Journal, Computers & electrical engineering.
- **Springer**: Multimedia Tools and Applications
- **Other**: ICCAIS2023, ICCAIS 2024, ICCAS 2024, ICCAS 2025, IEEE SIMPAR2025, All Life, Journal of Automation, Mobile Robotics and Intelligent Systems

IEEE , Student Member Graduate	2024-present
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IEEE Robotics and Automation Society , Student Member Graduate	2024-present
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IEEE Young Professionals , Student Member Graduate	2024-present
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REFERENCES

- Prof. Dr. [Nak Young Chong](#), Ph.D. (Ph.D. Supervisor)
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Director of the JAIST Robotics Laboratory
Co-founder and CEO of CURA Robotics and AI
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- A/Prof. Dr. [Xiem HoangVan](#), Ph.D (Undergraduate Supervisor)
Professor, University of Engineering and Technology, Vietnam National University, Vietnam.
Head of Robotics Engineering Department
Email: xiemhoang@vnu.edu.vn; Phone: (+84)378608113